Biochemical analysis of wound fluid from nonhealing and healing chronic leg ulcers

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The purpose of this study was to determine the biochemical composition of fluid taken from chronic wounds, to compare these values with that of serum, and therefore to assess whether the wound fluid is representative of the extracellular environment of the wound. Paired wound fluid and blood samples were collected from eight patients with chronic leg ulcers in a nonhealing and healing phase. Wound fluid and serum samples were screened for a profile of general biochemical analyses, including electrolytes, lactate, glucose, and protein analyses. Electrolyte levels were essentially identical in wound fluid and serum samples. Lactate and lactate dehydrogenase levels were significantly greater and glucose and bicarbonate levels were significantly lower in wound fluid when compared with the paired serum samples. Albumin and total protein levels in wound fluid were on average half those of serum levels. In this small sample of eight patients, wound fluid collected from chronic leg ulcers is an exudate with the biochemical composition expected in extracellular fluid. In addition, bicarbonate and glucose levels increase and C-reactive protein levels decrease in wound fluid, but remain unchanged in serum, during healing. These results suggest changes in the state of hypoxia and the inflammatory process in the healing wound. **(WOUND REP REG 1996;4:234-9)**

Currently, there is considerable clinical research into the treatment of chronic venous leg ulcers. However, the underlying pathophysiologic characteristics of the delayed healing process in these chronic wounds is largely unknown.[1,2]

Recently, the analysis of wound fluid has been used to gain an insight into the wound environment.[3,4] This research is based on the assumption that fluid collected from the wound surface represents extracellular fluid from the wound itself. If this is the case, then analysis of wound fluid may give valuable information about the growth conditions of cells within the wound. Collection methods have been variable and unstandardized and may involve collecting fluid from polyvinyl alcohol sponges and wire mesh chambers as well as fluid that has been sitting in a wound drainage bottle for many hours.[3,5,6] It is possible that at the time of analysis this fluid may bear little relationship to extracellular fluid.

There has been no detailed biochemical analysis of wound fluid to characterize its composition and to draw a comparison with serum. Wound fluid remains an undefined "soup" that bathes the cells in the wound environment. In addition, there has been no demonstration of biochemical changes which might occur in wound fluid from chronic leg ulcers when they are

ALT	Alanine aminotransferase
AST	Aspartate aminotransferase
C3	Complement factor 3
C4	Complement factor 4
CK	Creatine kinase
GGT	Gamma glutamyltranspeptidase
LDH	Lactate dehydrogenase
Tbil	Total bilirubin

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actively healing. Such information would help provide a better understanding of the healing process in chronic wounds.

It was therefore the purpose of this study to compare wound fluid, collected in a standardized manner, with serum to understand whether wound fluid truly represents extracellular fluid. This study also aims to identify changes in the biochemical composition of wound fluid collected from chronic leg ulcers at nonhealing and healing phases.

MATERIALS AND METHODS

Patient sample

All patients examined in this study had chronic leg ulcers on the lower leg excluding the foot. The cause of each ulcer was assessed by careful history, examination, and evaluation by a series of investigations. The investigations performed were venous refilling time by photoplethysmography,[7] arterial Doppler pressures,[8] and a panel of blood tests that included standard serum biochemistry analysis, urea and electrolytes, liver function tests, glucose levels, and rheumatoid serologic analysis. A diagnosis of venous disease was confirmed by a venous refilling time less than 25 seconds and arterial disease was determined by an ankle/brachial Doppler arterial ratio of less than 0.9. All patients, who were considered to have venous disease, had been treated with compression therapy with two compression bandages (Bieirsdorf, Hamburg, Germany) and Tubigrip (Seton, Oldham, United Kingdom) to improve venous function. In all cases the patients' ulcers had failed to respond to outpatient treatment as defined by no reduction in the size of the ulcer in more than 3 months or a continued increase in the size of the ulcer. Patients were admitted to the hospital for bed rest, 6 hourly saline solution compresses, and eventual skin grafting. The admission did not include intervention for other potential causes of delayed healing. Informed consent was obtained from each patient, and ethics approval was given by the Fremantle Hospital and University of Western Australia ethics committees.

Collection of wound fluid and serum samples

Wound fluid and blood samples were collected from patients within 24 hours of admission to the hospital (nonhealing phase) and after 2 weeks of regular dressings and bed rest (healing phase). Wound fluid was collected from each patient in a standardized manner. The patient was fasted beginning at midnight, and a transparent occlusive film (Opsite dressing; Smith & Nephew, Clayton, Australia) was placed over the wound at 8:00 AM. The patient's leg was placed in a dependent position, and the patient was encouraged to drink 1 L of water, which aided in standardizing the state of hydration of the patient. The fluid was aspirated from beneath the dressing after a period of 1 hour and then transferred into Greiner Vacuette vacuum serum collection tubes (Interpath Ltd, Melbourne, Australia) for all tests except lactate for which fluoride oxalate tubes were used. Serum samples were collected directly into the same type of vacuum collection tubes used for the wound fluid collection.

Ulcer assessment

The surface area of all ulcers was measured on admission and at 2 weeks by tracing around the edge of the ulcer onto clear plastic and then photocopying the tracing onto paper. The surface area of the ulcer was then calculated with the use of a Kent digital planimeter (Jayco, Sydney, Australia). These measurements were used to calculate the percentage reduction in size of the ulcer after 2 weeks as an indication that healing was occurring. All ulcers were also photographed on admission and after 2 weeks to provide a visual record of any changes in appearance of the ulcer, such as reduction of slough, granulation tissue formation, and epithelialization of the wound edge.

Biochemical analysis

Wound fluid and serum samples were analyzed for a wide range of general biochemical parameters and measured with standard automated biochemical analyzers. Osmolality was measured with an osmometer (model 3MO; Advanced Instruments, Norwood, Mass.). Sodium, potassium, chloride, bicarbonate, calcium, magnesium, phosphate, glucose, lactate, lactate dehydrogenase (LDH), alkaline phosphatase, alanine aminotransferase (ALT), aspartate aminotransferase (AST), gamma glutamyltranspeptidase (GGT), total bilirubin (Tbil), creatine kinase (CK), urea, creatinine, uric acid, total protein, albumin, cholesterol and triglycerides were all measured with the Kodak Ektachem 700 automated biochemical analyser (Eastman Kodak, Rochester, N.Y.). C-reactive protein, complement factor 3 (C3), and complement factor 4 (C4) were estimated with the Behringwerke Turbitimer (Behringwerke, Marburg, Germany). Alpha-1-globulin, alpha-2-globulin, beta-globulin, and gamma-globulin were measured with a Helena Electrophoresis System (Helena Laboratories, Beaumont, Tex.). The accepted serum reference ranges for each parameter are included in Table 1.

Table 1. Comparison of biochemical analytes for all serum and wound fluid samples

Biochemical Test	Serum reference range	Serum		Wound Fluid		*p* Value*
		Median	Range	Median	Range	
Osmolality	(275-295 mOsmol)	**286**	271-308	**287**	257-305	NS
Sodium	(137-146 mmol/L)	**142**	138-145	**141**	133-146	NS
Potassium	(3.6-5.0 mmol/L)	**4.9**	3.6-6.6	**4.4**	3.2-5.7	<0.01
Chloride	(98-108 mmol/L)	**101**	95-106	**104**	96-109	<0.01
Urea	(2.0-8.6 mmol/L)	**8.7**	2.7-23.1	**8.9**	2.5-22.6	NS
Creatinine	(45-115 μmol/L)	**112**	59-342	**117**	46-334	NS
Uric acid	(140-480 μmol/L)	**418**	236-823	**347**	221-751	<0.05
Calcium†	(2.15-2.5 mmol/L)	**2.29**	2.21-2.46	**2.23**	1.84-2.72	NS
Magnesium	(0.7-0.91 mmol/L)	**0.89**	0.69-1.31	**0.94**	0.75-1.24	NS
Phosphate	(0.8-1.46 mmol/L)	**1.16**	1.02-1.68	**1.27**	0.98-1.47	NS
Bicarbonate	(22-32 mmol/L)	**26**	19-31	**19**	14-22	<0.001
Glucose	(3.5-5.5 mmol/L)	**6.4**	4.6-9.9	**1.8**	0.6-5.9	<0.001
Lactate	(0.3-1.3 mmol/L)	**2.7**	1.3-6.5	**10.9**	5.4-16.7	<0.01
LDH	(310-620 U/L)	**381**	275-567	**5378**	789-9901	<0.001
Alkaline phosphatase	(35-120 U/L)	**120**	58-247	**84**	24-146	<0.01
ALT	(8-60 U/L)	**26**	13-64	**24**	3-111	NS
AST	(4-50 U/L)	**30**	11-81	**61**	10-159	<0.01
GGT	(8-60 U/L)	**61**	23-432	**46**	19-194	<0.05
CK	(20-260 U/L)	**54**	20-155	**29**	20-62	<0.05
Total bilirubin	(3-20 μmol/L)	**13**	12-35	**22**	12-52	<0.05
Total protein	(56-79 gm/L)	**73**	65-83	**39**	26-51	<0.001
Albumin	(36-50 gm/l)	**40**	35-48	**22**	14-28	<0.001
C-reactive protein	(<6 mg/L)	**6.4**	5-115	**8.5**	2.5-25	NS
α-1-Globulin	(2-5 gm/L)	**3.0**	1.4-4.4	**1.4**	1.0-2.5	<0.001
α-2-Globulin	(4-9 gm/L)	**9.2**	5.2-11.8	**4.4**	2.7-6.0	<0.001
β-Globulin	(5-10 gm/L)	**11**	8.6-13.7	**5.8**	3.1-8.8	<0.001
γ-Globulin	(8-16 gm/L)	**9.8**	7.9-13.9	**5.7**	3.9-9.0	<0.001
C3	(0.55-1.2 gm/L)	**0.95**	0.49-1.15	**0.46**	0.15-0.71	<0.001
C4	(0.2-0.5 gm/L)	**0.37**	0.2-0.66	**0.22**	0.08-0.35	<0.001
Cholesterol	(5.5 mmol/L)	**4.4**	1.1-7.0	**1.7**	1.2-3.2	<0.001
Triglycerides	(0.3-1.8 mmol/L)	**1.9**	0.8-3.7	**0.6**	0.4-4.3	<0.01

NS, Not significant.
*Significance level for comparison of wound fluid value to paired serum value
†Calcium = albumin adjusted Calcium

Microbiology

Qualitative bacteriologic characteristics were determined by taking swabs of the ulcer which were then cultured with selective media.

Quantitative bacteriologic analysis was also carried out on the wound fluid samples. Wound fluid was thoroughly mixed, and a 10 μl aliquot of wound fluid was serially diluted (1:20, 1:200, 1:2000). Dilutions were plated onto blood agar plates, and the number of colonies on each plate counted and the number of bacteria per milliliter of wound fluid was calculated.

Statistical analysis

Median values were calculated and paired samples were compared statistically by means of the Wilcoxon sign-rank test with the Statistical Package for Social Sciences (SPSS).

RESULTS

Wound fluid and serum samples were collected from eight patients with chronic leg ulcers of the gaiter region in both nonhealing and healing phases. There were six male and two female patients, with a median age of 73 years (range 69 to 89 years). In four patients the ulcers were due to venous disease alone, and in four patients the cause was venous and arterial disease. The Doppler ratios of patients with arterial disease ranged from 0.45 to 0.7. Two patients with venous disease had non-insulin-dependent diabetes. These patients did not receive additional treatment for the arterial component of their disease or any alteration in their diabetic medications after admission to the hospital.

The median initial size of the ulcers was 68 cm^2 with a range of 50 to 198 cm^2. The median reduction

Table 2. Comparison of selected biochemical analytes from nonhealing and healing wound fluids

	Nonhealing		Healing	
Biochemical Test	**Median**	**Range**	**Median**	**Range**
Biocarbonate (mmol/L)	**17.5**	14-20	**19***	16-22
Glucose (mmol/L)	**1.2**	0.6-3.7	**2***	1.1-5.9
Total protein (gm/L)	**34**	26-46	**41†**	36-51
Albumin (gm/L)	**19**	14-24	**23†**	18-28
C-reactive protein (mg/L)	**13**	5-25	**5***	2.5-21
γ-Globulin (gm/L)	**4.5**	3.9-6.6	**6†**	4.4-9.0
Cholesterol (mmol/L)	**1.6**	1.2-3.2	**1.8‡**	1.3-3.2

Each "healing" wound fluid value is significantly different from "nonhealing" fluid value at the following level of significance.
* $p = 0.02$
† $p = 0.01$
‡ $p = 0.05$

Table 3. Comparison of selected biochemical analytes from nonhealing and healing serum

	Nonhealing		Healing	
Biochemical Test	**Median**	**Range**	**Median**	**Range**
Bicarbonate (mmol/L)	**27**	19-31	**26***	20-30
Glucose (mmol/L)	**6.4**	4.6-9.9	**6.5***	5-9.6
Total protein (gm/L)	**70**	65-83	**77†**	69-81
Albumin (gm/L)	**38.5**	35-48	**42†**	39-47
C-reactive protein (mg/L)	**8**	5-115	**5.5***	5-45
γ-Globulin (gm/L)	**5.7**	3.9-9.0	**4.5***	3.9-6.6
Cholesterol (mmol/L)	**4.3**	1.1-6.0	**4.6***	3.9-7.0

Statistical comparison of serum from individuals during the healing and nonhealing phase.
* Not significant.
† $p = 0.02$.

in the size of the ulcers was 12% with a range of 1% to 23%. All ulcers showed signs of granulation tissue formation and epithelialization after the 2 weeks of hospitalization.

When comparing wound fluid with serum it was noted that there was no significant difference in sodium, magnesium, phosphate, urea, and creatinine levels between wound fluid and serum samples ($p > 0.05$) (Table 1). The concentration of other basic electrolyte components (potassium and chloride) were different in wound fluid compared with serum. However, the levels were within the normal range for serum and consequently that expected in extracellular fluid. When corrected for the lower albumin level, no difference was found in calcium concentration.[9] The osmolality of serum and wound fluid were not significantly different.

Lactate levels were significantly increased in wound fluid (median 10.9 mmol/L) relative to serum (2.7 mmol/L) ($p < 0.01$), and glucose levels were significantly lower in wound fluid (1.8 mmol/L) when compared with paired serum samples (6.4 mmol/L) ($p < 0.01$). In addition, the levels of bicarbonate were significantly lower in wound fluid (19 mmol/L) when compared with serum (26 mmol/L) ($p < 0.001$). These findings are consistent with the existence of an acidotic, anaerobic environment. The levels of LDH in wound fluid (5378 μg/L) were almost 20 times those in serum (381 μg/L). Other cellular enzymes were present in lower concentrations in wound fluid (alkaline phosphatase, CK, GGT), were unchanged (ALT), or were slightly elevated (AST).

Both albumin and total protein levels in wound fluid (21 and 38.5 gm/L, respectively) were approximately half the levels present in serum (40 and 73 gm/L). This trend was consistent for alpha, beta, and gamma globulin concentrations as determined by electrophoresis. Complement factors C3 and C4 were also present in lower levels in wound fluid. However, the levels of C-reactive protein were similar in wound fluid (8.5 mg/L) and serum (6.4 mg/L) ($p > 0.05$). Cholesterol and triglyceride concentrations were also reduced in wound fluid relative to serum, which is consistent with the lower protein levels.[10] Additional findings were a higher Tbil content and lower uric acid level in wound fluid.

When we analyzed changes in the wound fluid from the nonhealing to the healing phase (Table 2), it was noted that there was a significant increase in the levels of bicarbonate, glucose, albumin, and total protein, gamma-globulin, and cholesterol. In contrast, there was a significant decrease in the levels of C-reactive protein. However, when comparing the results of the analysis of the nonhealing and healing serum (Table 3), it can be seen that there is a concurrent increase in the serum levels of total protein and albumin from the nonhealing to the healing phase, whereas bicarbonate, glucose, C-reactive protein gamma-globulin, and cholesterol remain unaltered. All other parameters remained unchanged in wound fluid or serum (data not shown).

The results of the qualitative bacteriologic analysis showed that all ulcers cultured bacteria at both phases. The bacteria detected in the ulcers were *Staphylococcus aureus*, hemolytic Streptococci group G, *Pseudomonas aeruginosa*, mixed coliforms, and skin flora. Analysis of the wound fluid for quantitative levels of bacteria was carried out on seven of the eight patients. The median number of bacteria detected in initial samples was 1.76×10^6 with a range of 1×10^4 to 2.2×10^7. Samples from four patients showed a decrease in the number of bacteria with healing, and three samples showed an increase in the number of bacteria in the wound fluid in the healing phase.

DISCUSSION

The analysis of fluid collected from acute and chronic human wounds is being increasingly used as an investigative tool in the expanding area of wound healing research.[11-14] Interest in the study of wound fluid is based on the assumption that it is an exudate representing the extracellular fluid of the tissue and as such may provide valuable information on changes occurring in the wound environment. It may be assumed that this fluid contains many factors that can influence cell growth and that by analyzing wound fluid an understanding of changes occurring in the wound may be gained.

There are many different ways of collecting wound fluid each of which may influence the composition of the fluid being analyzed. These methods have included blister fluid, fluid from beneath transparent films and wound drainage bottles. It is therefore important to know that the wound fluid has a biochemical composition equivalent to extracellular fluid and has not been significantly altered by the collection process. The wound fluid in this study has the electrolyte composition, urea, and creatinine levels and also the osmolality equivalent to serum and hence appears to reflect the extracellular environment of the wound.

On the basis of comparative studies, biologic fluids such as pleural effusion and ascites may be categorized as exudates or transudates. An exudate can be defined as a fluid deposited in tissues (as a result of inflammation or infection) that has high viscosity and high protein (>30 gm/L). A transudate is a fluid that has passed through a membrane and has high fluidity and low protein (<30 gm/L).[15-17] The results of the protein analysis in this study (median protein 38.5 gm/L) indicate that wound fluid collected from chronic leg ulcers in the manner described is an exudate. This finding is also supported by the levels of albumin and cholesterol. The increase in protein and albumin with healing of the chronic wounds may simply reflect improved nutritional and constitutional status with hospitalization because this change is present both in serum and wound fluid. The initial serum levels of albumin and protein do not represent malnutrition, but this increase in concentration with healing may indicate that improved nutritional status could be beneficial in the healing of some wounds. The increase in protein and albumin levels may also be a result of greater amino acid mobilization because of the enforced bed rest.

The high levels of lactate and low levels of bicarbonate and glucose in wound fluid when compared with serum suggest an anaerobic environment or altered state of metabolism within the wound. The subsequent increase in bicarbonate and glucose content associated with healing ulcers suggests that the wound hypoxia may be reduced in the healing phase. This is indirect evidence of such changes but is consistent with the theoretic pathogenesis of chronic leg ulcers caused by venous and arterial disease.

The unexpectedly high levels of LDH may be indicative of a number of pathologic conditions. First, this may be a direct result of the high lactate production within the wound. LDH is the enzyme responsible for the conversion of lactate to pyruvate and may be induced to compensate for the high levels of lactate present. However, LDH is also a marker of widespread tissue necrosis, for example, in myocardial infarction or liver disease, and as such could be indicative of a high level of tissue damage.[18-21] However, in the absence of any comparable increase in other cellular enzymes this is not likely to be the case.

The changes observed in C-reactive protein although small were in direct contrast to the other proteins measured, in that there is no difference in the levels of C-reactive protein in the total wound fluid compared with serum; but there is a significant decrease in the C-reactive protein levels in wound fluid during healing, as opposed to the marked increase in total protein and albumin levels. The changes noted in C-reactive protein were particularly noteworthy in that this protein is often used as a clinical marker of inflammation, for example, in disorders such as rheumatoid arthritis, pancreatitis, postoperative complications, appendicitis, and infection.[22-27] Therefore the decrease in the levels of C-reactive protein from the nonhealing to the healing phase suggests a decreased inflammatory state in the healing wound. This hypothesis lends further support to the concept that nonhealing chronic leg ulcers represent a chronic inflammatory disease process and as such require a treatment approach appropriate to addressing the problem of chronic inflammation.[28] Whether the C-reactive protein detected in wound fluid is simply being deposited at this site in response to the inflammation and tissue destruction or whether it is being produced locally within the wound remains open to conjecture.

The changes observed in wound fluid with healing of the leg ulcers could be due to factors other than an alteration in the healing process. A reduced bacterial contamination of the ulcers as a result of hospitalization could result in diminished stimulation by inflammatory mediators and hence inflammation. However, we did not show an alteration in the type of bacteria or a reduction in the number of bacteria in the wound fluid with healing of the ulcers. This find-

ing leads to the conclusion that the alterations in bacterial contamination are not responsible for the changes observed. However, a larger sample size and quantification in wound biopsies would help to show more clearly whether bacteria are responsible for the observed changes.

The serum and wound fluid biochemical analysis results suggest that wound fluid represents extracellular fluid of the wound and as such is a useful tool for analyzing the wound environment. Analysis of standard clinical biochemical parameters analytes within wound fluid has not been previously reported and as shown here can give information about the metabolic state of the wound. By extending this analysis to examine both nonhealing and healing chronic wounds, valuable information can be gained about changes occurring in the wound during healing.

This study was performed on a small number of patients and as such is limited; however, all significant changes were observed consistently in all patients, giving strong support to the suggestion that the changes occurring are due to the healing process. Further studies need to be performed on a larger number of patients to confirm that the changes are due to wound healing and not simply an artifact of the small patient numbers or due to other factors associated with hospitalization.

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